

Belief Explorer: A Preliminary Evaluation of AI-Mediated Socratic Dialogue for Epistemic Reflection

Manuel Delaflor
Metacognition Institute
Chesterfield, Derbyshire, United
Kingdom
manuel@metacognitioninstitute.org

Cecilia Delgado Solorzano
School of Computing
Clemson University
Clemson, South Carolina, USA
cecilid@clemson.edu

Carlos Toxtli
Human-AI Empowerment Lab
Clemson University
Clemson, South Carolina, USA
ctoxkli@clemson.edu

Abstract

AI chatbots may reinforce existing beliefs and discourage critical examination. This study evaluates Belief Explorer, an AI system that uses Socratic dialogue and multi-perspective analysis to support epistemic reflection. Participants recruited via Prolific used Belief Explorer to examine personally held beliefs in contested domains (e.g., climate change, origin of life). Post-intervention surveys measured system usability, perceived epistemic impact, and self-reported reflection. Participants responded positively: a large majority reported that Belief Explorer felt substantially different from conventional AI chatbots, prompted deep reflection on their beliefs, and increased awareness of underlying assumptions. Thematic analysis of open-ended responses revealed that participants valued the tool's non-judgmental approach, multi-perspective framing, and analytical feedback mechanisms. Belief change itself was modest and bidirectional, with similar proportions reporting increased and decreased confidence in their initial beliefs. This preliminary study offers proof-of-concept that AI systems with epistemic scaffolding can support critical thinking and self-reflection. Future research should use controlled experimental designs to establish causal relationships and examine long-term effects on epistemic attitudes.

CCS Concepts

• **General and reference** → Evaluation; • **Human-centered computing** → *Empirical studies in HCI*; • **Applied computing** → **Sociology**; **Interactive learning environments**; • **Theory of computation** → **Semantics and reasoning**; • **Computing methodologies** → **Cognitive science**; **Theory of mind**.

Keywords

epistemic reflection, AI chatbots, Socratic dialogue, critical thinking, belief examination, human-AI interaction

ACM Reference Format:

Manuel Delaflor, Cecilia Delgado Solorzano, and Carlos Toxtli. 2026. Belief Explorer: A Preliminary Evaluation of AI-Mediated Socratic Dialogue for Epistemic Reflection. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3772363.3799391>



This work is licensed under a Creative Commons Attribution 4.0 International License. *CHI EA '26, Barcelona, Spain*

© 2026 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2281-3/26/04

<https://doi.org/10.1145/3772363.3799391>

1 Introduction

Large language model (LLM)-based AI assistants have changed how people access and process information. These tools offer convenience, but they may also reinforce confirmation bias and discourage critical examination of beliefs Bail [1]. Traditional search engines present multiple sources for user evaluation; conversational AI systems typically provide confident, singular responses that can discourage deeper inquiry. Recent research confirms this concern: when challenged with counterarguments, LLMs like ChatGPT frequently abandon correct positions, suggesting they lack stable epistemic grounding Wang et al. [9]. The problem intensifies in domains with contested claims, where different worldviews and evidentiary standards produce incompatible conclusions. In these contexts, definitive answers matter less than understanding the epistemic foundations of competing perspectives: what counts as evidence, how claims get contextualized, and where assumptions diverge. Belief Explorer takes a different approach to AI-mediated information interaction. The system uses Socratic questioning to guide users in examining their own beliefs, presents multi-perspective analyses that reveal epistemic tensions, and offers quantitative metrics that make the contested nature of claims explicit. The system scaffolds epistemic reflection, meaning the metacognitive process of examining one's own beliefs, evidence evaluation practices, and underlying assumptions.

1.1 Research Questions

This preliminary study addresses the following research questions:

RQ1: Do users report increased epistemic reflection after engaging with Belief Explorer?

RQ2: How do users perceive Belief Explorer compared to conventional AI assistants?

RQ3: What features of the system do users find most valuable for belief examination?

RQ4: Does engagement with Belief Explorer influence belief confidence or perspective-taking?

2 Related Work

Epistemic reflection refers to the metacognitive process of examining one's own knowledge claims, belief formation processes, and evidentiary standards Kuhn and Weinstock [5]. Research in cognitive psychology shows that individuals rarely engage in such reflection spontaneously, especially when beliefs are identity-relevant or emotionally salient Kahan [4]. Effective epistemic interventions therefore require explicit scaffolding that prompts individuals to articulate their reasoning, consider alternative perspectives, and evaluate evidentiary support. The Socratic method uses targeted

questions to expose contradictions and guide discovery, and has proven effective in promoting deeper reasoning across educational contexts. AI implementations of Socratic pedagogy show promise in scientific reasoning and moral philosophy Betz et al. [2], though applications to personally held beliefs remain limited.

Recent work has explored AI systems for critical thinking and perspective diversity. Tanprasert et al. examined debate chatbots on YouTube, finding that social identity and conversational style affect engagement with opposing views Tanprasert et al. [8]. Zhang et al. proposed a multi-agent system to burst filter bubbles through diverse viewpoint exposure Zhang et al. [10]. Belief Explorer differs in its epistemological grounding: rather than simply exposing users to opposing views, it operationalizes Model-Dependent Ontology (MDO) through multi-arbiter evaluation, making explicit that competing frameworks can produce valid but incompatible assessments of the same claim.

MDO evaluates beliefs by their pragmatic utility and predictive capacity within specific domains, setting aside questions of correspondence to metaphysical truth Delaflor [3]. MDO posits that all human understanding consists of simplified representations constrained to particular contexts, and that models gain acceptance based on their effectiveness in describing and predicting observations within their intended domains. For contested beliefs, this perspective shifts the goal from adjudicating between frameworks toward understanding how different models structure evidence and produce divergent yet internally coherent accounts.

3 Method

3.1 Design

This study used a single-group post-test design with mixed methods. Participants used Belief Explorer and then completed a survey measuring system usability, self-reported epistemic changes, and comparative evaluations. Open-ended responses provided qualitative insights into user experiences.

3.2 Participants

Thirty-one participants were recruited via Prolific between December 29, 2025, and January 7, 2026. One participant discontinued participation after 8% survey completion, resulting in a final sample of $N = 30$ (16 male, 14 female). Educational attainment varied: 14 bachelor's, 6 master's, 5 some college, 4 associate, 1 doctoral, 1 high school. Participants received standard Prolific compensation.

3.3 Belief Explorer System

Belief Explorer is a web-based AI system that supports epistemic reflection through three core components (Figure 1): **Socratic Dialogue Interface**. Users engage in text-based conversation with an AI agent that uses targeted questions to elicit reasoning, identify assumptions, and explore implications of beliefs. **Multi-Perspective Analysis**. The system presents competing perspectives on claims, including scientific consensus, alternative interpretations, and contextual considerations. **Analytical Feedback Metrics**. Users receive quantitative indicators including: (a) Verifact Score, measuring empirical evidence strength; (b) radar chart visualizations showing Empirical Verifiability, Logical Consistency, and Falsifiability dimensions; (c) Contextual Sensitivity Index (CSI), showing how

the utility of a claim varies across different use contexts (e.g., a claim useful for policy decisions may be less useful for personal health choices); (d) Model Diversity Quotient, indicating degree of disagreement among analytical perspectives.

3.4 Procedure

Participants accessed Belief Explorer through a web link. They were instructed to explore a personally held belief in a contested domain (examples provided: shape of the Earth, climate change, origin of life). The system guided them through a Socratic dialogue, presented analytical feedback, and displayed multi-perspective analyses. After the interaction, participants completed Likert-scale items and open-ended questions. Median completion time was 49 minutes (range: 21-110 minutes).

3.5 Measures

All Likert-scale items used 5-point scales (Strongly Disagree to Strongly Agree unless otherwise noted). **System Metrics Validation (SMV; 4 items)**. Assessed perceived utility of analytical features. **Reflection and Engagement (RE; 3 items)**. Measured self-reported reflection. **System Comparison (SC; 3 items)**. Evaluated differentiation from conventional AI tools. **Epistemic Impact (EI; 3 items)**. Assessed reported changes in thinking. **Belief Change Assessment (BCA; 2 items)**. Changes in belief confidence and nature of influence. **Open-Ended Questions (OEQ; 3 items)**. Most valuable insight, differences from other AI, suggested improvements.

3.6 Analysis

Open-ended responses were analyzed using thematic analysis. Two researchers independently reviewed responses, generated initial codes, and iteratively refined themes through discussion until consensus was reached.

4 Results

4.1 System Metrics Validation

Participants evaluated Belief Explorer's analytical features positively (Table 1). The majority agreed that the Verifact Score (80%, $n = 24$) and radar charts (67%, $n = 20$) helped them understand empirical support and claim structure. The Model Diversity Quotient received the strongest endorsement (73%, $n = 22$) for highlighting claim complexity. The Contextual Sensitivity Index received more mixed responses, with 30% ($n = 9$) neutral.

4.2 Reflection and Engagement

Participants reported substantial epistemic reflection (Table 1). The majority agreed that Belief Explorer promoted deep reflection (83%, $n = 25$), increased awareness of underlying assumptions (73%, $n = 22$), and helped identify logical inconsistencies (70%, $n = 21$).

4.3 System Comparison

Participants clearly differentiated Belief Explorer from conventional AI assistants (Table 1). The vast majority agreed it provided a substantially different experience from ChatGPT (83%, $n = 25$), offered insights unavailable from search engines or standard AI (83%, $n =$

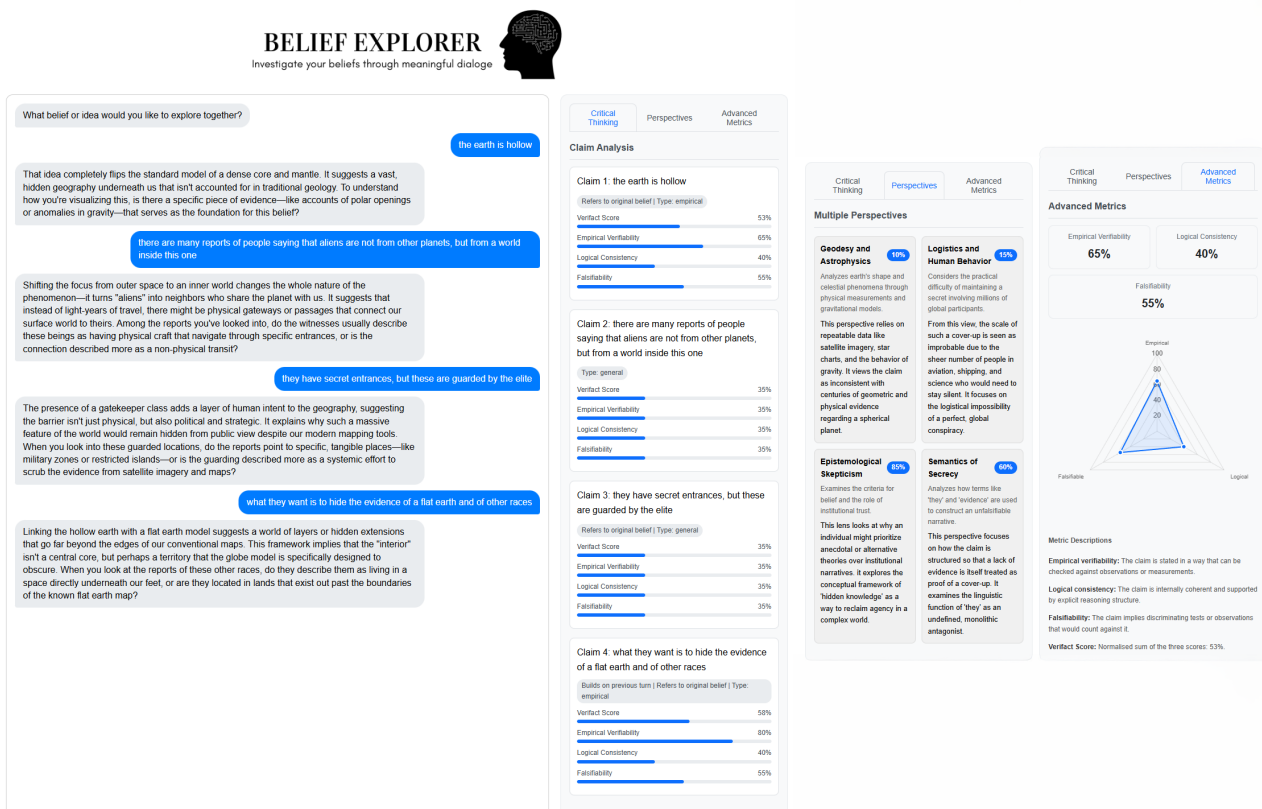


Figure 1: Belief Explorer interface. Left: Socratic dialogue where the AI poses probing questions rather than direct answers. Center: Claim Analysis panel showing per-claim metrics including Verifact Score, Empirical Verifiability, Logical Consistency, and Falsifiability. Right: Multiple Perspectives panel presenting diverse analytical frameworks with confidence scores, and Advanced Metrics panel displaying a radar visualization of epistemic dimensions.

Table 1: Survey Response Distribution

	Item	n	Median	IQR	% Disagree (1-2)	% Neutral (3)	% Agree (4-5)
0	SMV1	30	4	0	10	10	80
1	SMV2	30	4	1	16.7	16.7	66.7
2	SMV3	30	4	1	16.7	30	53.3
3	SMV4	30	4	1.75	13.3	13.3	73.3
4	RE1	30	4	1	10	6.7	83.3
5	RE2	30	4	1.75	10	16.7	73.3
6	RE3	30	4	2	16.7	13.3	70
7	SC1	30	4	1	6.7	10	83.3
8	SC2	30	4	1	10	6.7	83.3
9	SC3	30	4	0	3.3	10	86.7
10	EI1	30	4	1	20	16.7	63.3
11	EI2	30	4	1	3.3	20	76.7
12	EI3	30	4	1.75	16.7	20	63.3

%D = Disagree (1–2); %N = Neutral (3); %A = Agree (4–5). All items rated on a 5-point Likert scale (Strongly Disagree – Strongly Agree).

25), and featured a more effective Socratic approach than directive fact-checking (87%, $n = 26$).

4.4 Epistemic Impact

Self-reported epistemic changes were moderate (Table 1). A majority reported increased confidence in distinguishing well-supported from poorly-supported claims (77%, $n = 23$), greater likelihood of

questioning information before acceptance (63%, $n = 19$), and consideration of previously unconsidered perspectives (63%, $n = 19$). Substantial minorities reported no change, suggesting varied impact across participants.

4.5 Belief Change

Changes in belief confidence were bidirectional. While 40% ($n = 12$) reported increased confidence in their initial beliefs, 33% ($n = 10$) reported decreased confidence, and 27% ($n = 8$) reported no change. The most common response was that Belief Explorer introduced new perspectives worth considering (43%, $n = 13$), followed by refinement of existing perspectives (27%, $n = 8$). The full distribution of influence types is shown in Table 2.

Table 2: BCA2: Nature of Influence (N = 30)

Response	n	%
Introduced new perspectives now considering	13	43.3
Helped refine existing perspective	8	26.7
Reinforced what I already believed	4	13.3
Had no meaningful impact	3	10.0
Significantly changed my perspective	2	6.7

4.6 Qualitative Findings

Open-ended responses revealed three recurring themes:

Distinction Between Belief and Argument. Several participants noted that Belief Explorer helped them separate subjective beliefs from verifiable arguments. One participant wrote: ‘The most valuable insight I gained was the importance of distinguishing between belief and verifiable argument. In exploring my view on climate cycles, I realized I was conflating personal skepticism with actual scientific evidence.’

Non-Judgmental Exploration. Participants appreciated the system’s non-judgmental approach. One user noted: ‘Unlike other AI tools that immediately correct you, this one actually let me explore my belief without making me feel stupid. It asked questions that made me think about why I believe what I believe.’

Complexity Awareness. Several participants reported new appreciation for how contested topics lack simple answers. One participant stated: ‘The tool showed me that even within scientific consensus, there are debates about mechanisms, timescales, and policy implications.’

5 Discussion

This preliminary study provides encouraging evidence that AI systems can promote epistemic reflection. Participants overwhelmingly reported that Belief Explorer offered a qualitatively different experience from conventional AI assistants. The system’s Socratic approach and multi-perspective framing appear to have scaffolded metacognitive processes.

Belief Explorer’s architecture directly operationalizes MDO principles through its multi-arbiter analytical pipeline. Rather than positioning itself as an arbiter of truth, the system evaluates claims through three complementary lenses (Empirical, Logical, Pragmatic)

that reflect MDO’s emphasis on context-dependent utility. These arbiters frequently produce divergent assessments of the same claim, which is a feature from an MDO perspective, not a flaw. The Model Diversity Quotient makes this inter-arbiter disagreement visible, and the system presents competing evaluations in parallel rather than collapsing them into singular verdicts, inviting users to recognize that belief evaluation is inherently model-dependent Delaflor [3].

5.1 Implications

The strong differentiation from ChatGPT suggests that architectural choices matter, creating a qualitatively different interaction from conventional prompt-response patterns. The bidirectional changes in belief confidence suggest the system supports genuine inquiry and avoids directional persuasion. This pattern is consistent with intellectual humility frameworks, where uncertainty can indicate epistemic growth Leary et al. [6], Porter and Schumann [7].

The qualitative finding that participants valued distinguishing belief from argument deserves attention. This metacognitive insight, understanding that subjective conviction differs from evidentiary warrant, is a sophisticated epistemic stance often absent in online discourse. For educational applications, Belief Explorer shows AI can function as an intellectual sparring partner, with promise for political polarization, health misinformation, and scientific literacy.

5.2 Limitations

Several limitations qualify these findings. The single-group post-test design precludes causal claims; without control groups, we cannot definitively attribute reflection to the intervention. All measures rely on self-report, which may overestimate epistemic changes due to demand characteristics. The sample size ($N = 30$) and high educational attainment limit generalizability. We also lack data on belief durability and potential negative outcomes.

6 Conclusion

This preliminary evaluation shows that AI systems can promote epistemic reflection and intellectual humility. Belief Explorer’s combination of Socratic questioning, multi-perspective analysis, and analytical feedback appears to create conditions for examining beliefs without triggering defensiveness. The study’s design limits causal claims, but strong user endorsement and qualitative insights provide proof-of-concept justifying more rigorous investigation. As AI systems become embedded in information ecosystems, designing for epistemic responsibility becomes imperative. Belief Explorer offers one approach, prioritizing reflection over ready answers.

References

- [1] Christopher A Bail. 2024. Can Generative AI improve social science? *Proceedings of the National Academy of Sciences* 121, 21 (2024), e2314021121.
- [2] Gregor Betz, Christian Voigt, and Kyle Richardson. 2021. Critical Thinking for Language Models. In *Proceedings of the 14th International Conference on Computational Semantics (IWCS 2021)*. Association for Computational Linguistics, 63–75. <https://aclanthology.org/2021.iwcs-1.7>
- [3] Manuel Delaflor. 2024. Introduction to Model Dependent Ontology. doi:10.13140/RG.2.2.17190.11846
- [4] Dan M Kahan. 2017. Misconceptions, misinformation, and the logic of identity-protective cognition. (2017).
- [5] Deanna Kuhn and Michael Weinstock. 2012. What is epistemological thinking and why does it matter? In *Personal epistemology*. Routledge, 121–144.

- [6] Mark R Leary, Kate J Diebels, Erin K Davisson, Katrina P Jongman-Sereno, Jennifer C Isherwood, Kaitlin T Raimi, Samantha A Deffler, and Rick H Hoyle. 2017. Cognitive and interpersonal features of intellectual humility. *Personality and Social Psychology Bulletin* 43, 6 (2017), 793–813.
- [7] Tenelle Porter and Karina Schumann. 2018. Intellectual humility and openness to the opposing view. *Self and Identity* 17, 2 (2018), 139–162.
- [8] Tana Tanprasert, Sidney S. Fels, Luanne Sinnamon, and Dongwook Yoon. 2024. Debate Chatbots to Facilitate Critical Thinking on YouTube: Social Identity and Conversational Style Make a Difference. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*. ACM, 1–24.
- [9] Boshi Wang, Xiang Yue, and Huan Sun. 2023. Can ChatGPT Defend its Belief in Truth? Evaluating LLM Reasoning via Debate. In *Findings of the Association for Computational Linguistics: EMNLP 2023*. Association for Computational Linguistics, Singapore, 11865–11881. doi:10.18653/v1/2023.findings-emnlp.795
- [10] Yanzhe Zhang, Jing Sun, Ling Feng, Cheng Yao, Mingming Fan, Luyao Zhang, and Yong Rui. 2024. See Widely, Think Wisely: Toward Designing a Generative Multi-Agent System to Burst Filter Bubbles. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*. ACM, 1–24.